

ITA:0501 Computer Vision

~~Assignment~~ ASSIGNMENT 1

CO:1 - Assessment tool:2 - Concept Map

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1. Autonomous Vehicle - Concept Map

Scene Capture (camera)

- ↳ Captures real-world images or videos
- ↳ Shows roads, pedestrians, signs

Image Preprocessing (pre-processing)

- ↳ Noise reduction
- ↳ colour adjustment
- ↳ Resize/normalize

Feature Extraction

- ↳ Edges & lines
- ↳ Shapes & textures
- ↳ colour/motion features

Object recognition, navigation and decision making

- ↳ Detects vehicles, bicycles etc.,
- ↳ understands environment and plans safe path

Overview of computer vision

- ↳ Helps machine see and understand things
- ↳ combines image processing AI and pattern recognition

Features of computer vision

- ↳ Image capture & processing
- ↳ Feature extraction
- ↳ Object recognition
- ↳ real-time processing

Digital image fundamentals

- ↳ Image = 2D array of pixels
- ↳ Pixel value = intensity
- ↳ types: grayscale/colour

Image sensing and acquisition

- ↳ cameras/sensors
- ↳ capture light and convert to electrical signal

Sampling and Quantization

- ↳ Sampling: dividing image into pixels
- ↳ Quantization: assign intensity values

Image Formation models

- ↳ How light forms an image
- ↳ Pinhole/camera model
- ↳ Affects brightness, contrast, perspective.

Applications

- ↳ Autonomous vehicles
- ↳ Surveillance
- ↳ Robotics
- ↳ Navigation

2. Medical Imaging - Concept Map

